

Movie Theater Seat Reservation System

Kayla Santiesteban

Luis Bueno

Tatiana Hernandez

Tenishia Reynolds

National University

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Team Assignment One

Dr. Amir Schur

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Part A: Team Project Selection

Reel Solutions has been engaged to develop the Movie Theater Seat Reservation System to improve the booking process and theater management. The project team will consist of:

- The Product Owner: Kayla Santiesteban
- The Scrum Master: Tenishia Reynolds
- Developer 1: Tatiana Hernandez
- Developer 2: Luis Bueno

Project Description

The Reel Solutions will develop a Movie Theater Seat Reservation System that simplifies the process of selecting and reserving seats for a movie screening. The system will allow customers to view available showtimes, see the seating layout for a selected screening, identify available and wheelchair accessible seats, and view ticket pricing before completing a reservation.

Customers will be able to reserve a seat, receive a booking confirmation, and cancel an existing reservation if needed. Theater managers will be able to manage movie titles, showtimes, ticket pricing, and theater room assignments. They will also be able to track reserved and available seats for each showing and mark screenings as sold out when no seats remain. These features ensure that screening information stays accurate and up to date while providing customers with a straightforward reservation experience.

Planned Java Implementation

Reel Solutions will implement a small portion of the MVP in Java. The planned portion will demonstrate seat availability, reservation creation, duplicate-reservation prevention, ticket-price calculation, and booking confirmation.

Project Vision

The Movie Theater Seat Reservation System aims to give customers a simple and efficient way to plan their movie experience. Customers can view available showtimes, review the seating layout, identify accessible seating, select an available seat, and see the total ticket price before confirming their reservation. For theater managers, the system provides a centralized way to manage movie titles, showtimes, ticket pricing, and theater room assignments. It also offers tools to monitor seat availability and mark screenings as sold out. Together, these functions create a system that is easy for customers to use and effective for theater staff managing daily operations.

The Preliminary Project Features of the system include:

1. Movie Showtimes: Customers can view all available movie showtimes to choose when to attend.
2. Theater Seating Layout: Customers can view a visual seating layout showing available and reserved seats.
3. Seat Selection: Customers can select an available seat and reserve it.

4. Reservation Validation: The system will prevent customers from reserving seats that are already taken or creating duplicate reservations.
5. Reservation Confirmation: Customers will receive confirmation after successfully reserving a seat.
6. Ticket Pricing: The system will display the total ticket price before the customer confirms the reservation.
7. Attendance Tracking: Theater managers can view the number of reserved and available seats for a showing.
8. Movie Catalog Management: Theater managers can manage movie information, showtimes, ticket pricing, and theater room assignments.
9. Sold-Out Screenings: Theater managers can mark screenings as sold out when no additional seats are available.
10. Accessible Seating: Wheelchair accessible seats will be clearly identified within the seating layout.
11. Reservation Cancellation: Customers can cancel an existing reservation.

Based on the project's initial scope, the system should have these preliminary requirements:

1. Allow customers to view all available movie showtimes.
2. Display the seating layout for the selected screening, with available and reserved seats.
3. Allow customers to select and reserve any seat that is available.
4. Prevent customers from selecting a seat that has been reserved or creating duplicate reservations.
5. Provide a booking confirmation after a reservation is completed.
6. Calculate and display the total ticket price before the reservation is finalized.
7. Allow theater managers to monitor reserved and available seats for each showing.
8. Allow theater managers to manage movie titles, showtimes, ticket pricing, and theater room assignments.
9. Allow theater managers to mark a screening as sold out.
10. Clearly highlight wheelchair accessible seats within the seating layout.
11. Allow customers to cancel an existing reservation.

Part B: Team Environment Setup

Reel Solutions established a shared GitHub environment to streamline coordination and maintain transparency during development. With the GitHub project board serving as a single hub for organizing tasks and monitoring progress, work can be managed efficiently and remain aligned throughout the project.

Repository: <https://github.com/TatianaH25/Movie-Theater-Seat-Reservation-System>

Project Board: <https://github.com/users/TatianaH25/projects/1>

Part C: Sprint Planning

Reel Solutions transformed the project's preliminary requirements into a focused set of user stories that will guide development of the Movie Theater Seat Reservation System. Built around the real needs of customers and theater managers, these stories capture the essential functions and provide clear direction for what must be delivered. They serve as the backbone of the project, shaping priorities and keeping development aligned with the intended user experience.

User Stories:

1. As a customer, I want to view available movie showtimes so I can choose when to attend.
2. As a customer, I want to see the available seats so I can choose where I want to sit.
3. As a customer, I want to reserve a seat that is available to secure my spot for the movie.
4. As a customer, I want the system to prevent duplicate or invalid reservations when selecting a seat.
5. As a customer, I want to receive a booking confirmation to have proof of my reservation.
6. As a customer, I want to see the total price of my ticket before confirming to know the final cost.
7. As a theatre manager, I want to see a total count of reserved and available seats to track attendance for a showing.
8. As a theatre manager, I need a way to manage the entire movie catalog, including showtimes, ticket pricing, and theatre room assignments.
9. As a theatre manager, I want to label a specific screening as "Sold Out" on the main menu, so customers are aware it is full and don't attempt to access it.
10. As a customer, I want the system to highlight wheelchair-accessible seats to easily find accommodations.
11. As a customer, I need to cancel my existing reservation.

Four-Week Sprint Plan

- Week 1: Gather requirements, finalize user scenarios, and organize the project backlog.
- Week 2: Create the architectural design, object model, and required UML diagrams.
- Week 3: Implement and test the selected Java portion of the reservation workflow.
- Week 4: Continue implementation, testing, correct defects, and complete the final report.